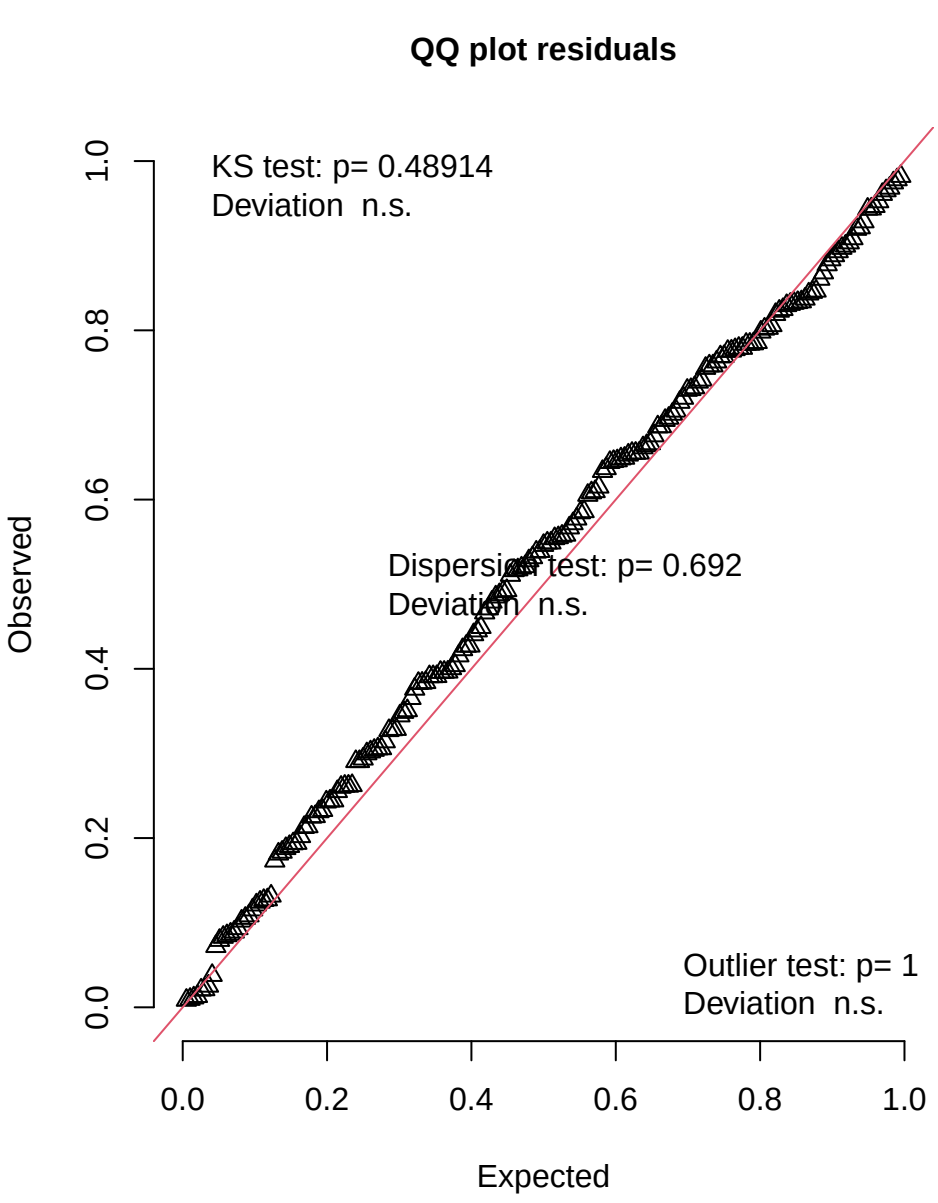
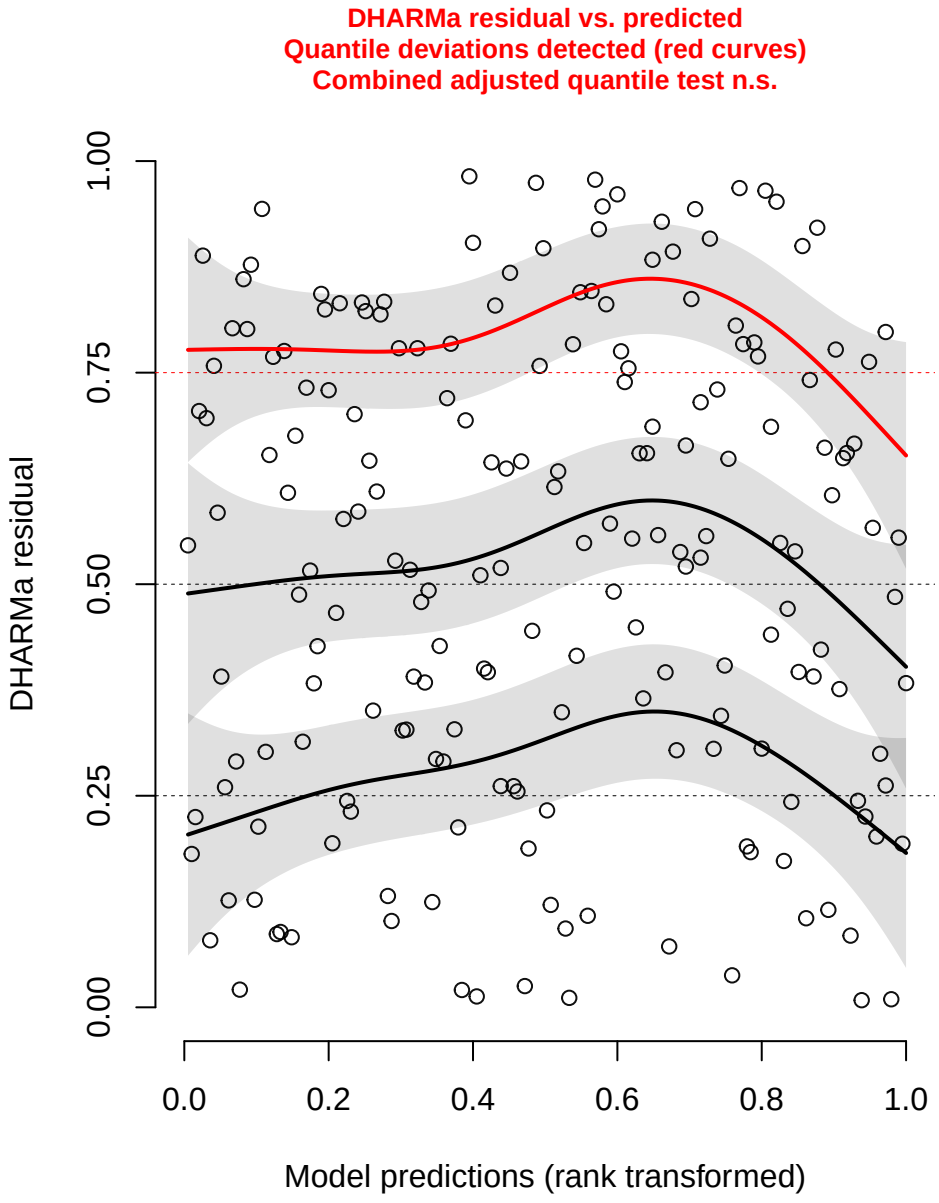
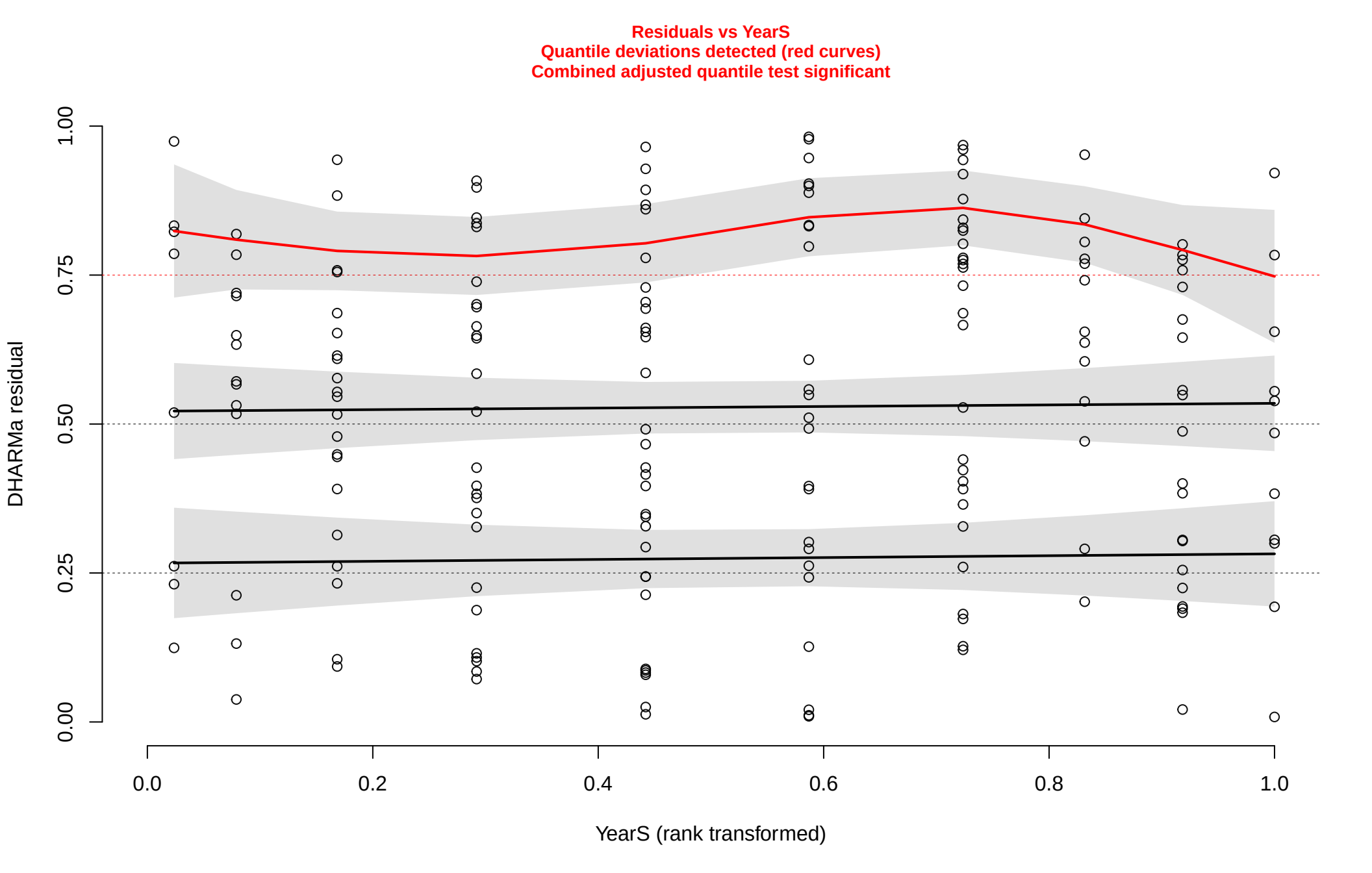
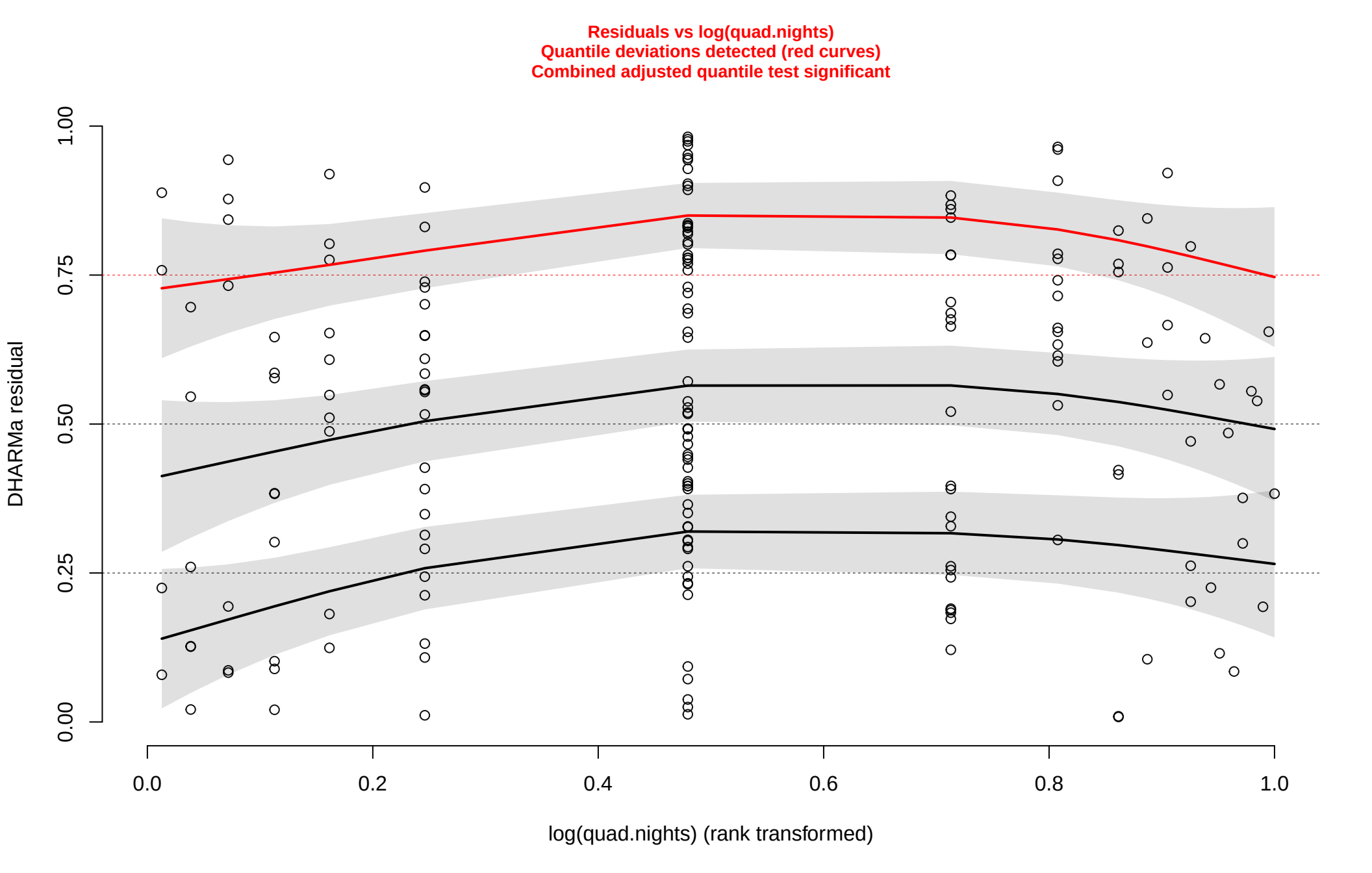
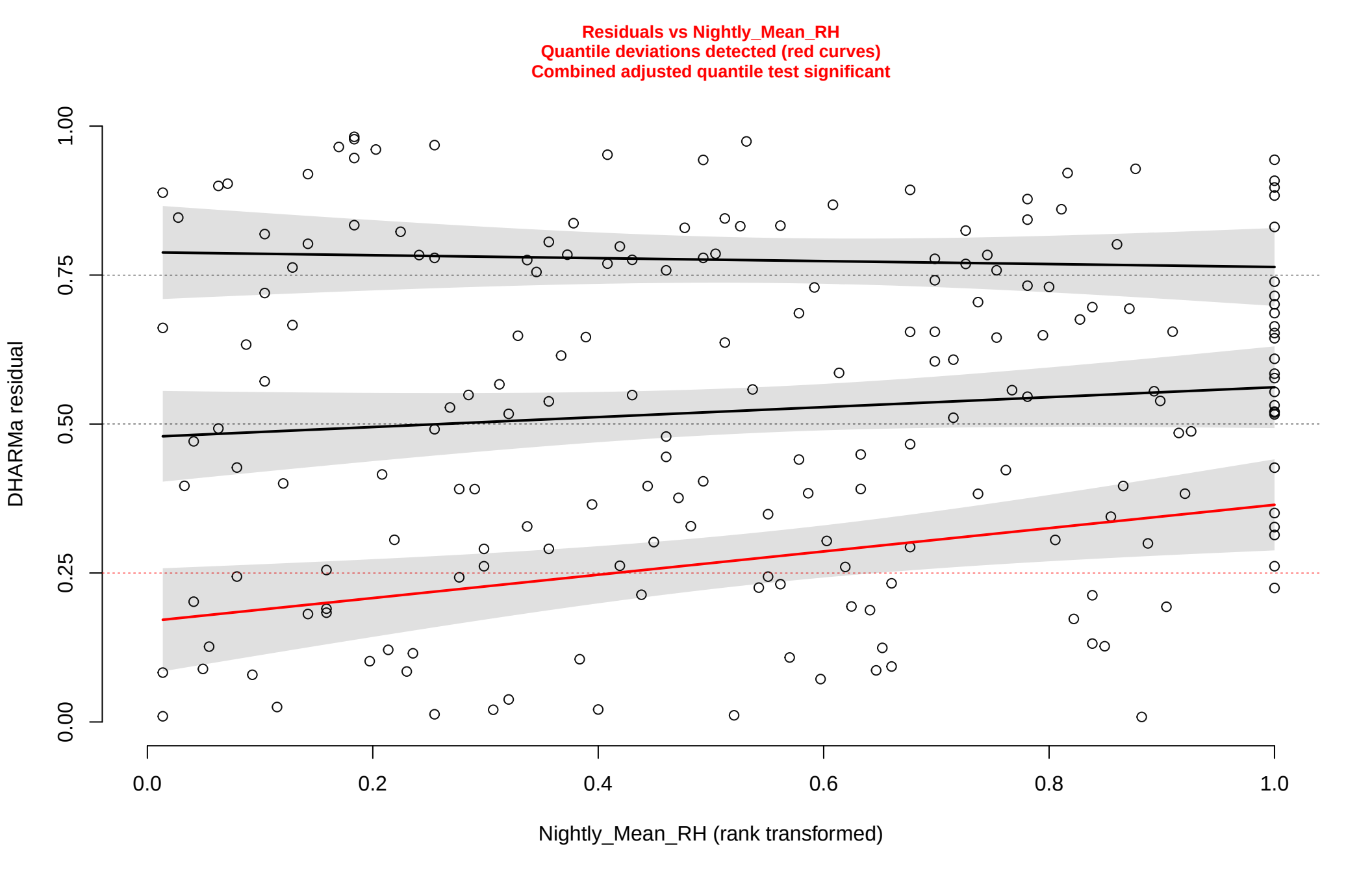
DHARMa residuals: Northern Region_LANO

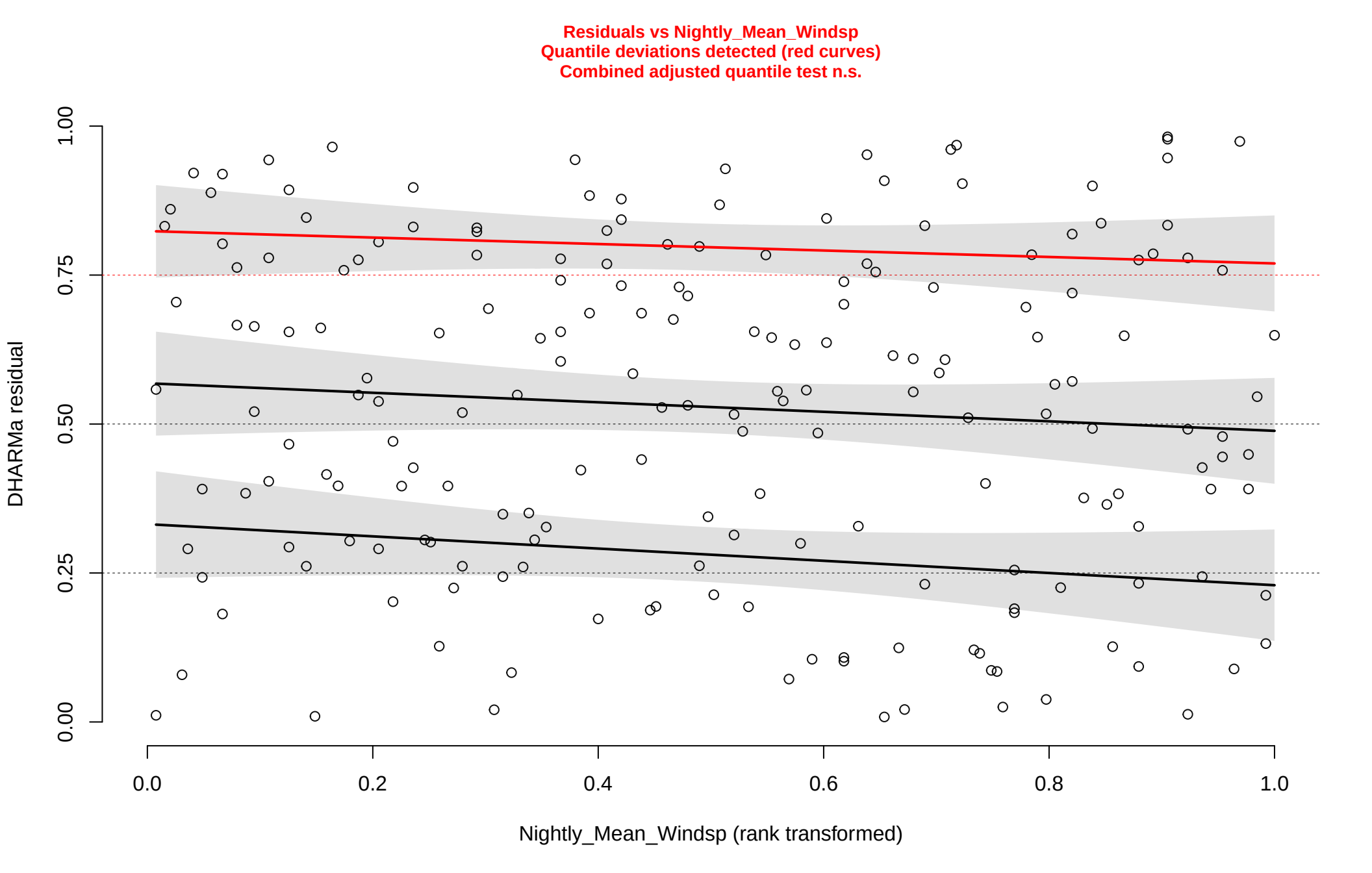


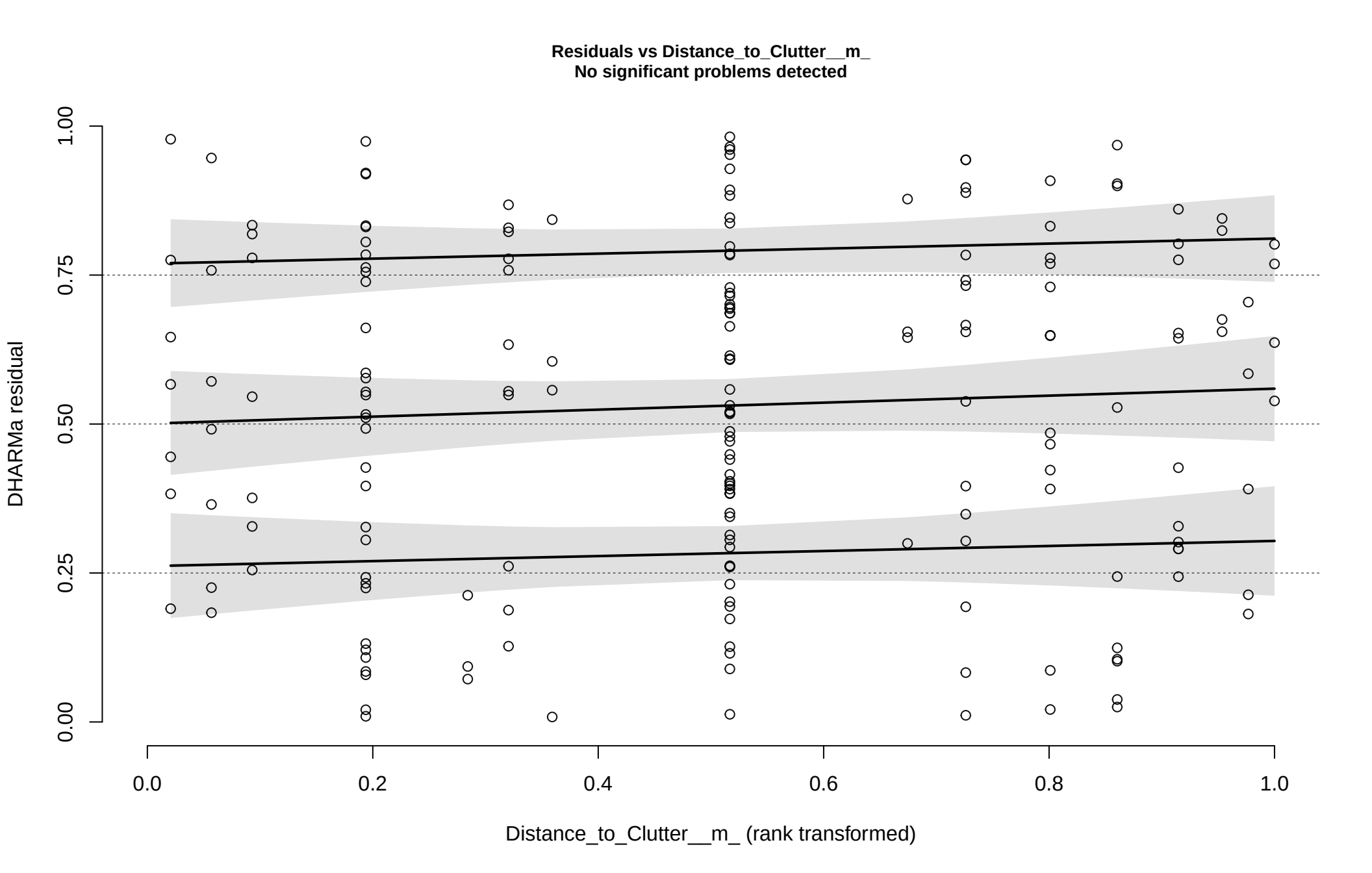


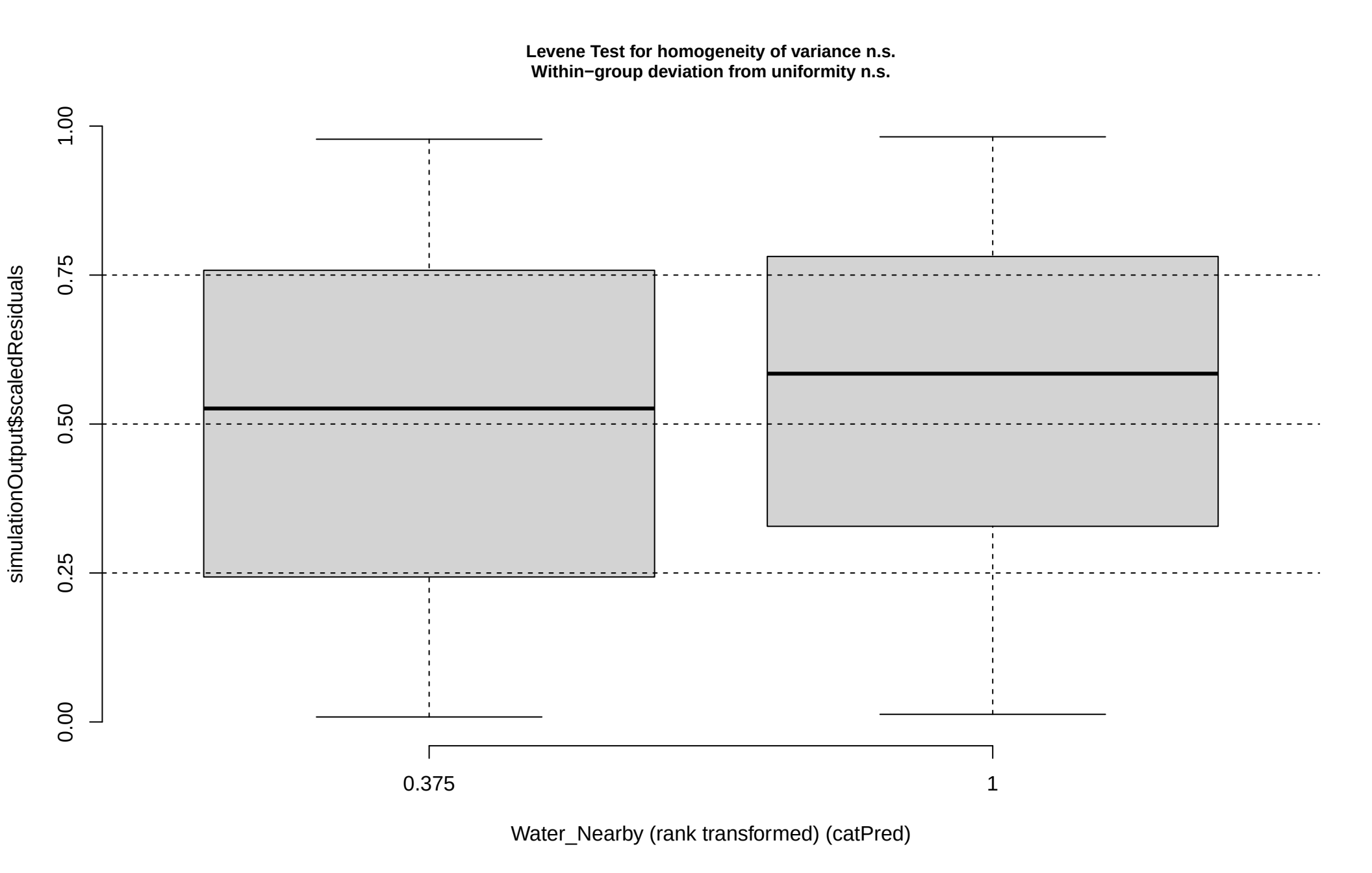

Residuals vs YearS
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

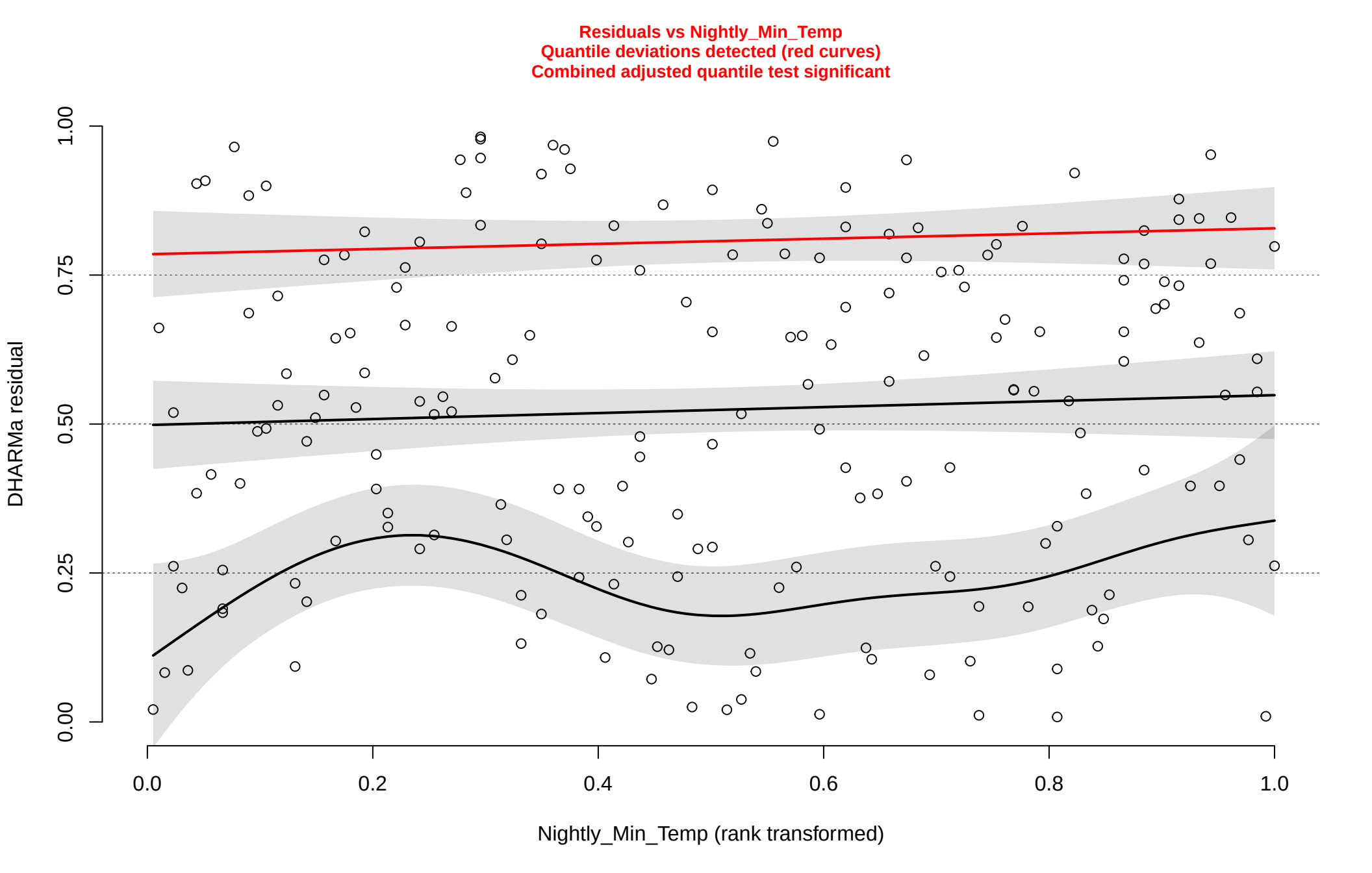

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

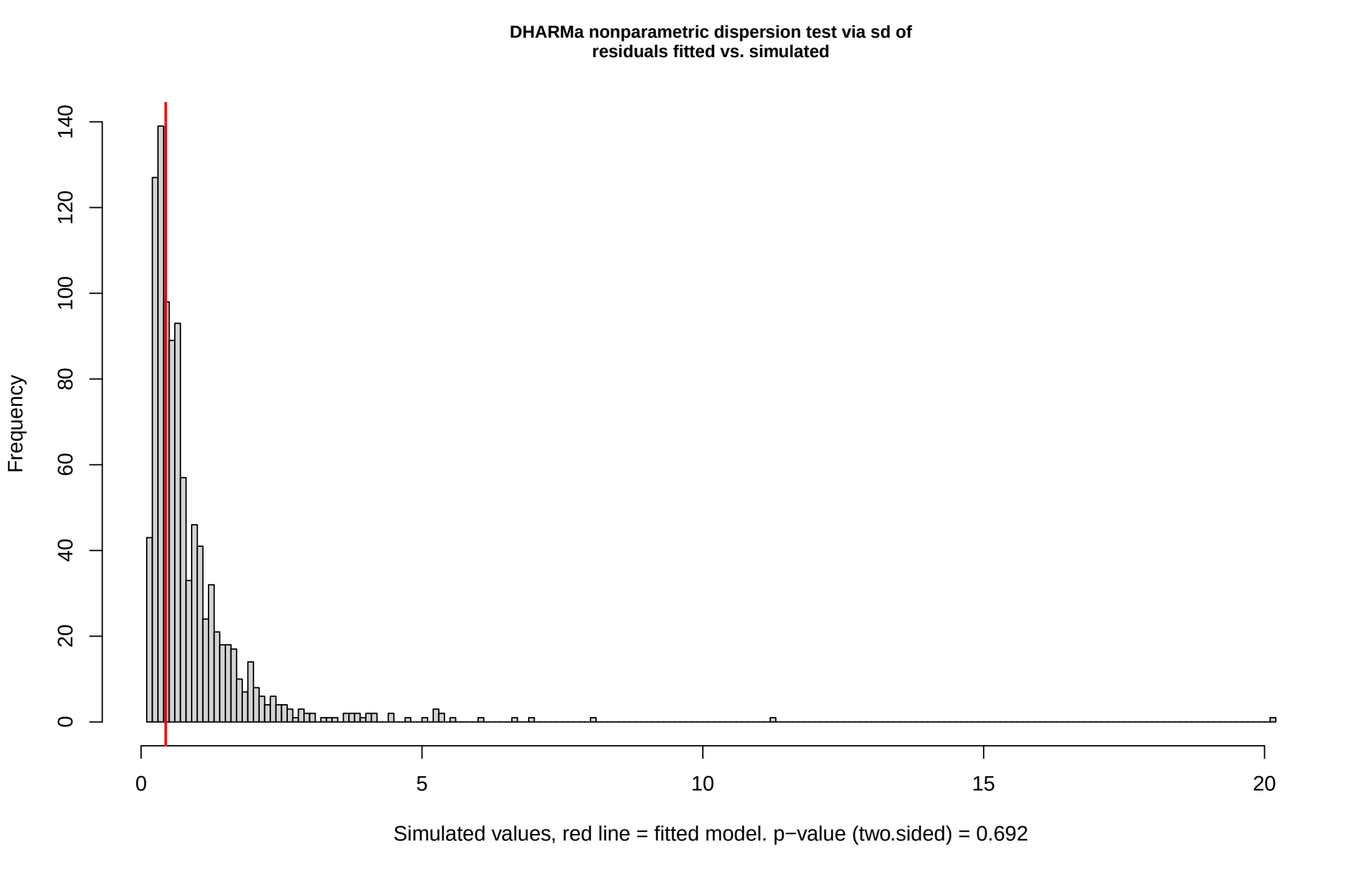

Residuals vs Nightly_Mean_RH
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_RH (rank transformed)

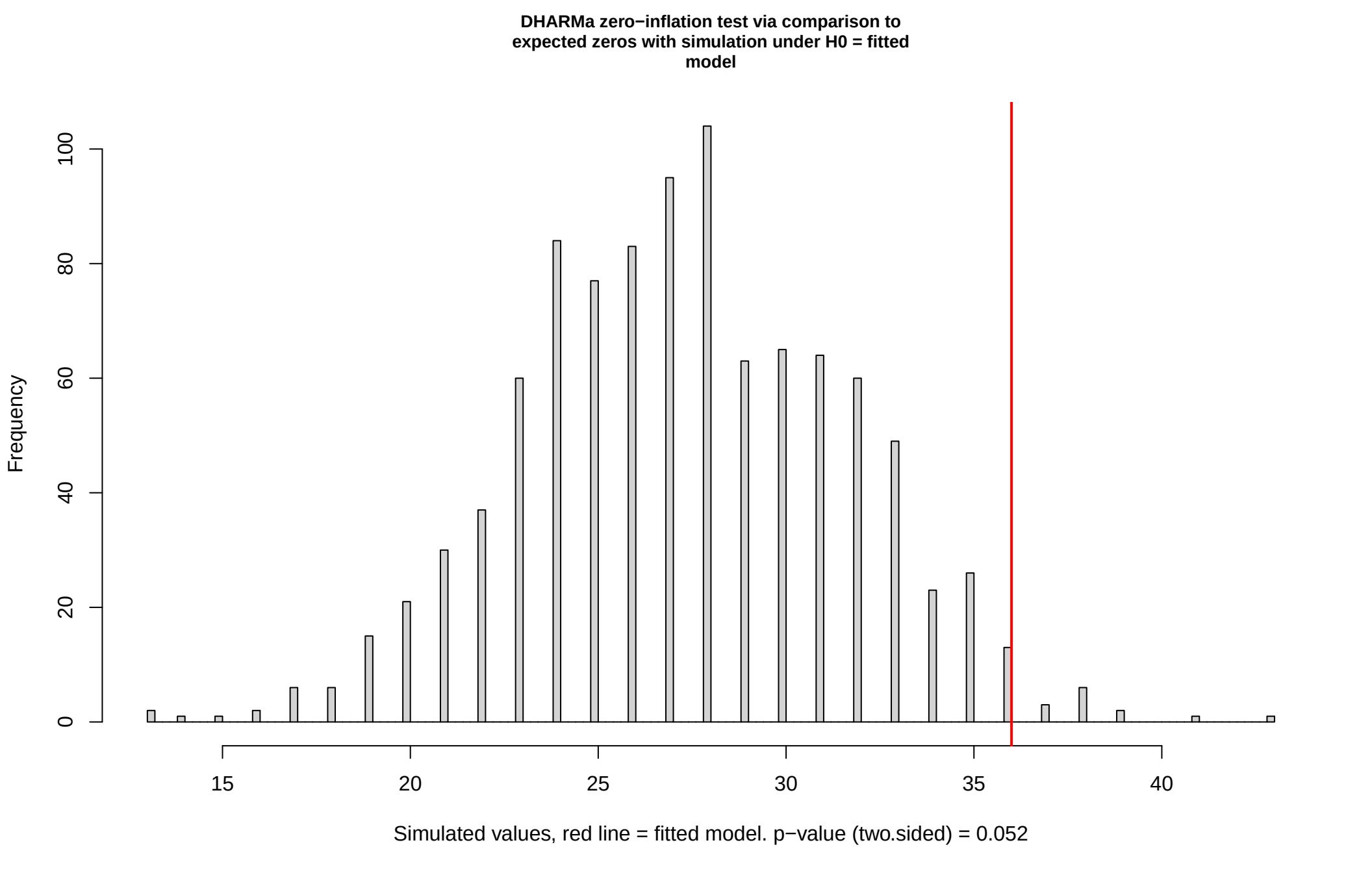

Residuals vs Nightly_Mean_Windsp
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Windsp (rank transformed)

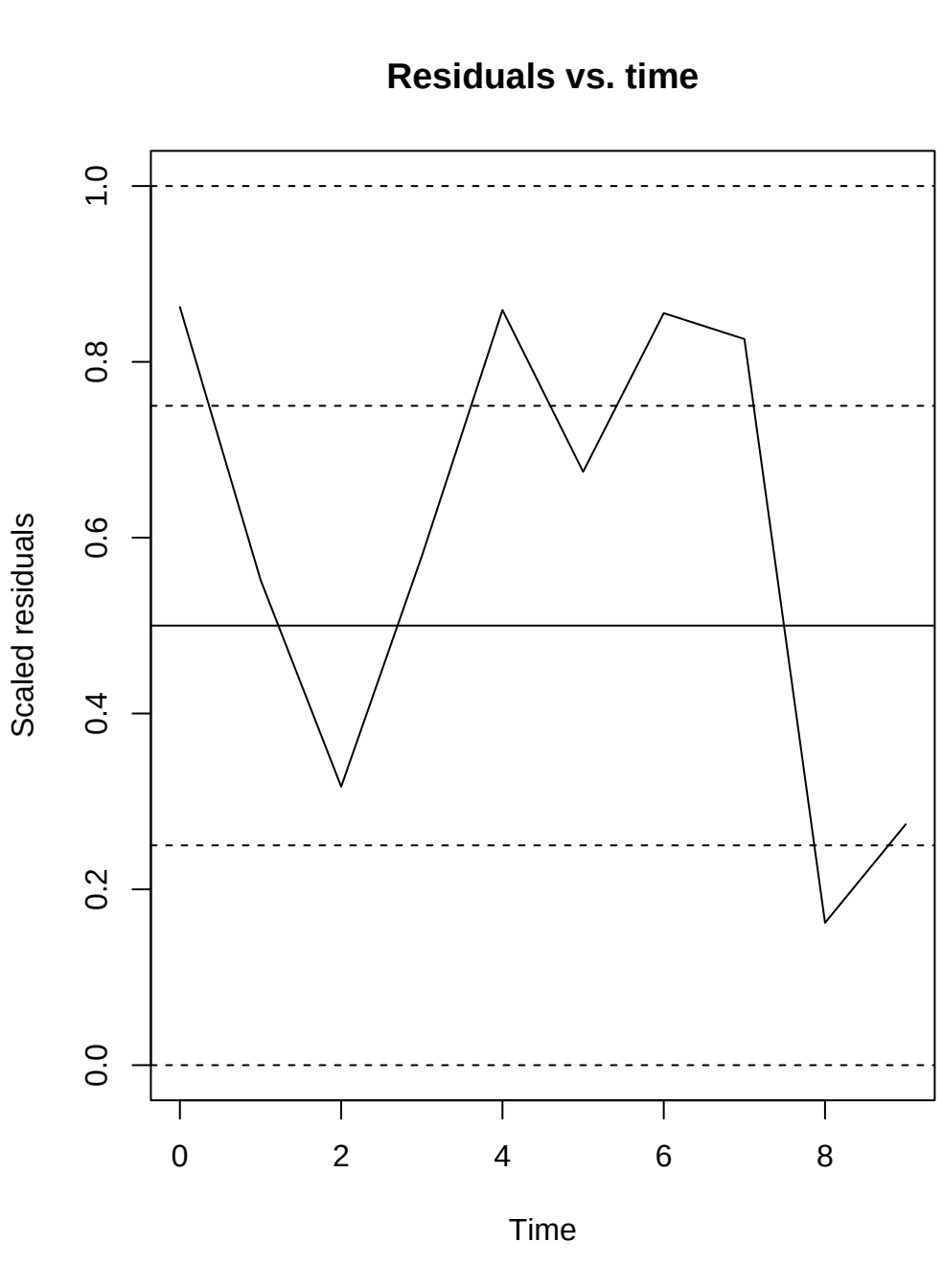

Residuals vs Distance_to_Clutter__m_
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

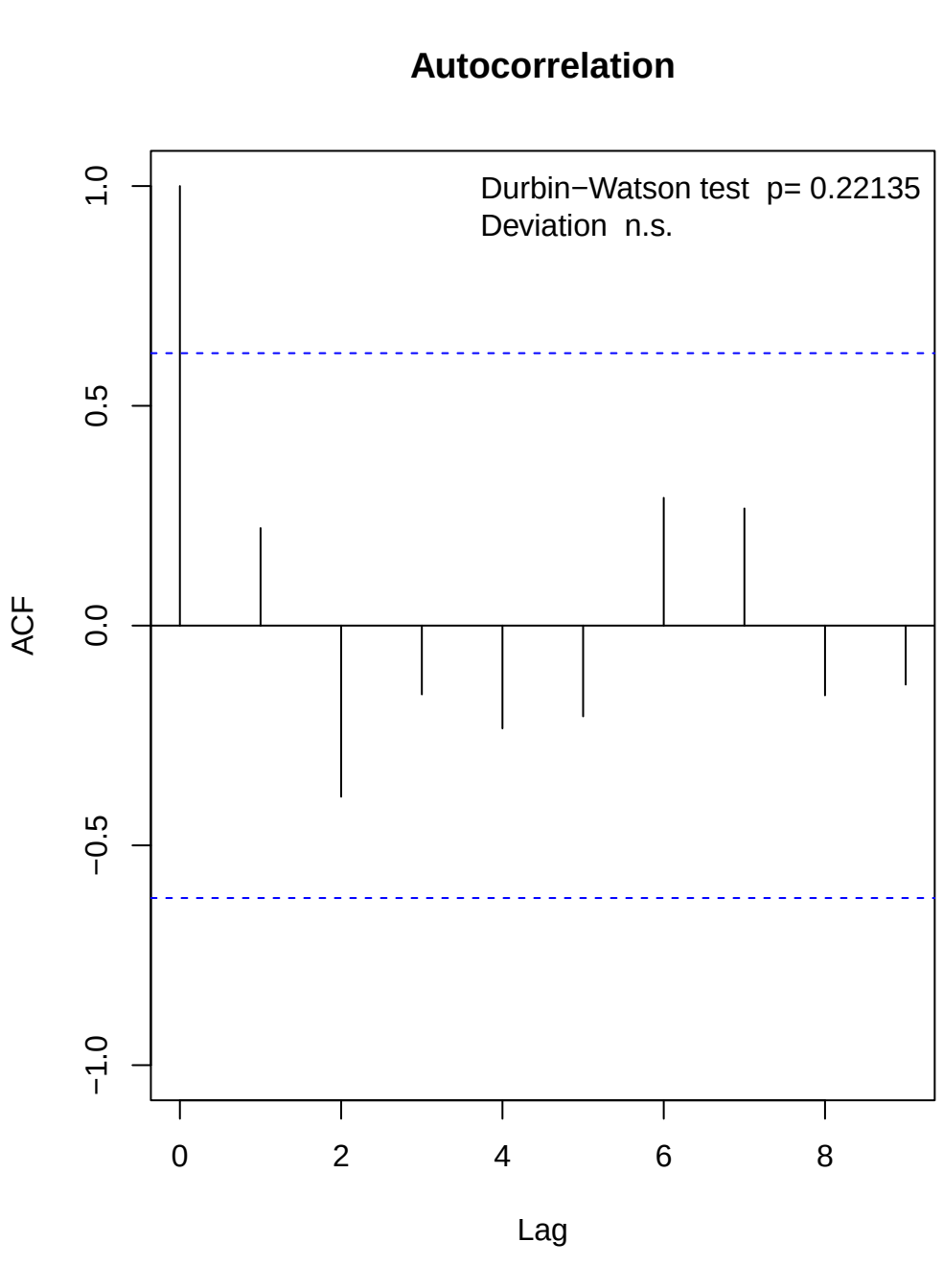

Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
0.375
1
Water_Nearby (rank transformed) (catPred)

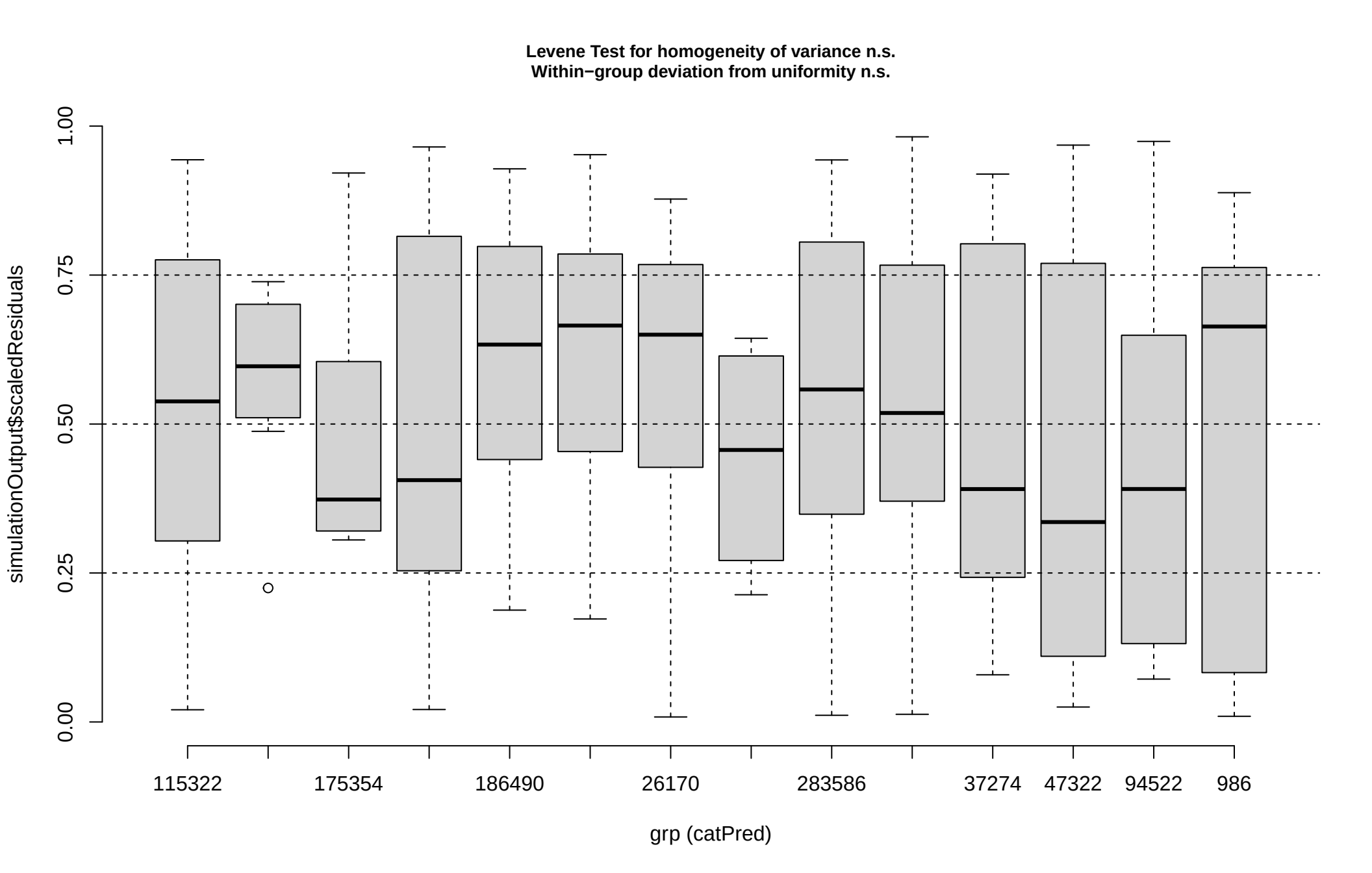

Residuals vs Nightly_Min_Temp
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Min_Temp (rank transformed)

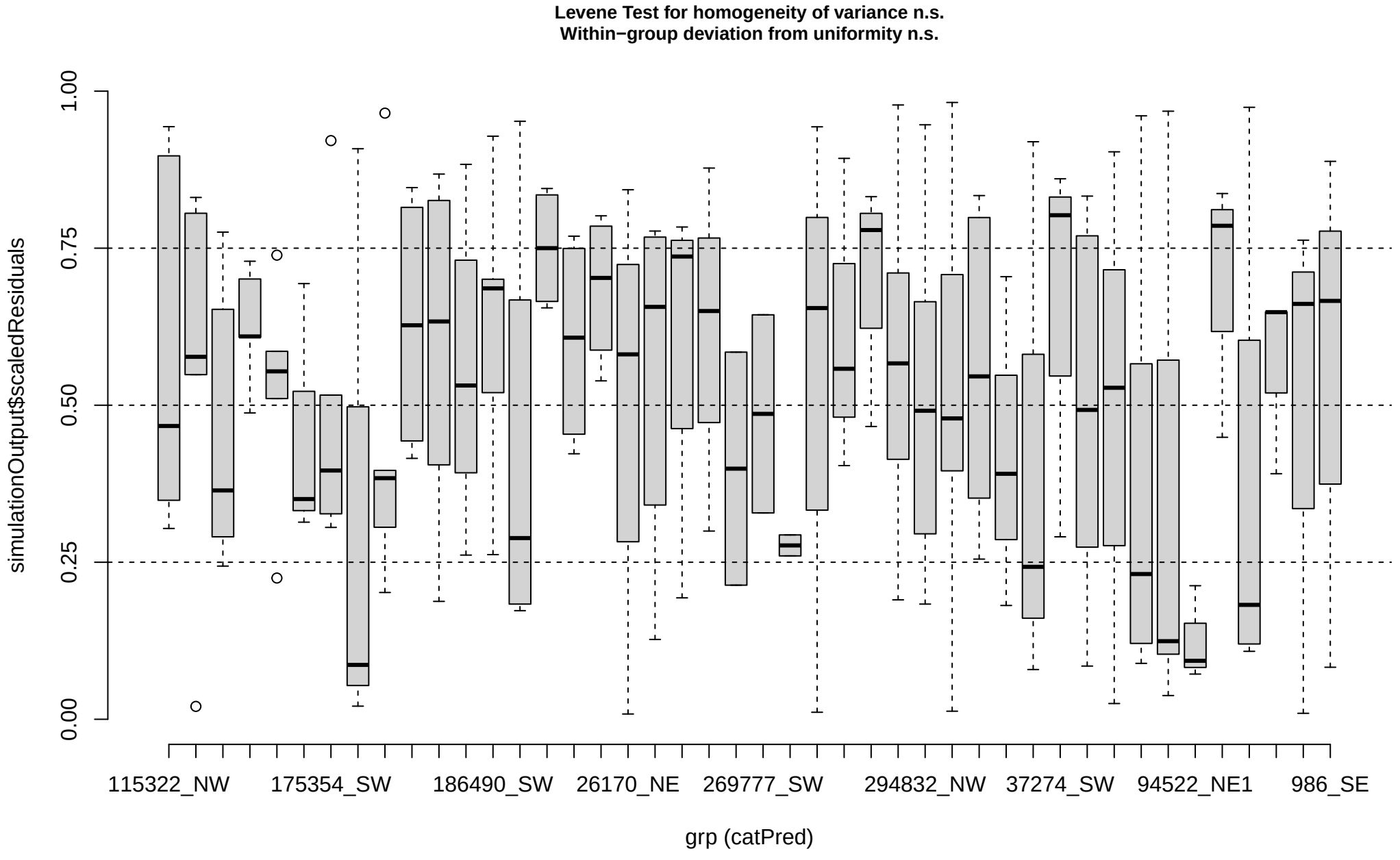

DHARMa nonparametric dispersion test via sd of
residuals fitted vs. simulated
Frequency
0
20
40
60
80
100
120
140
0
5
10
15
20
Simulated values, red line = fitted model. p−value (two.sided) = 0.692

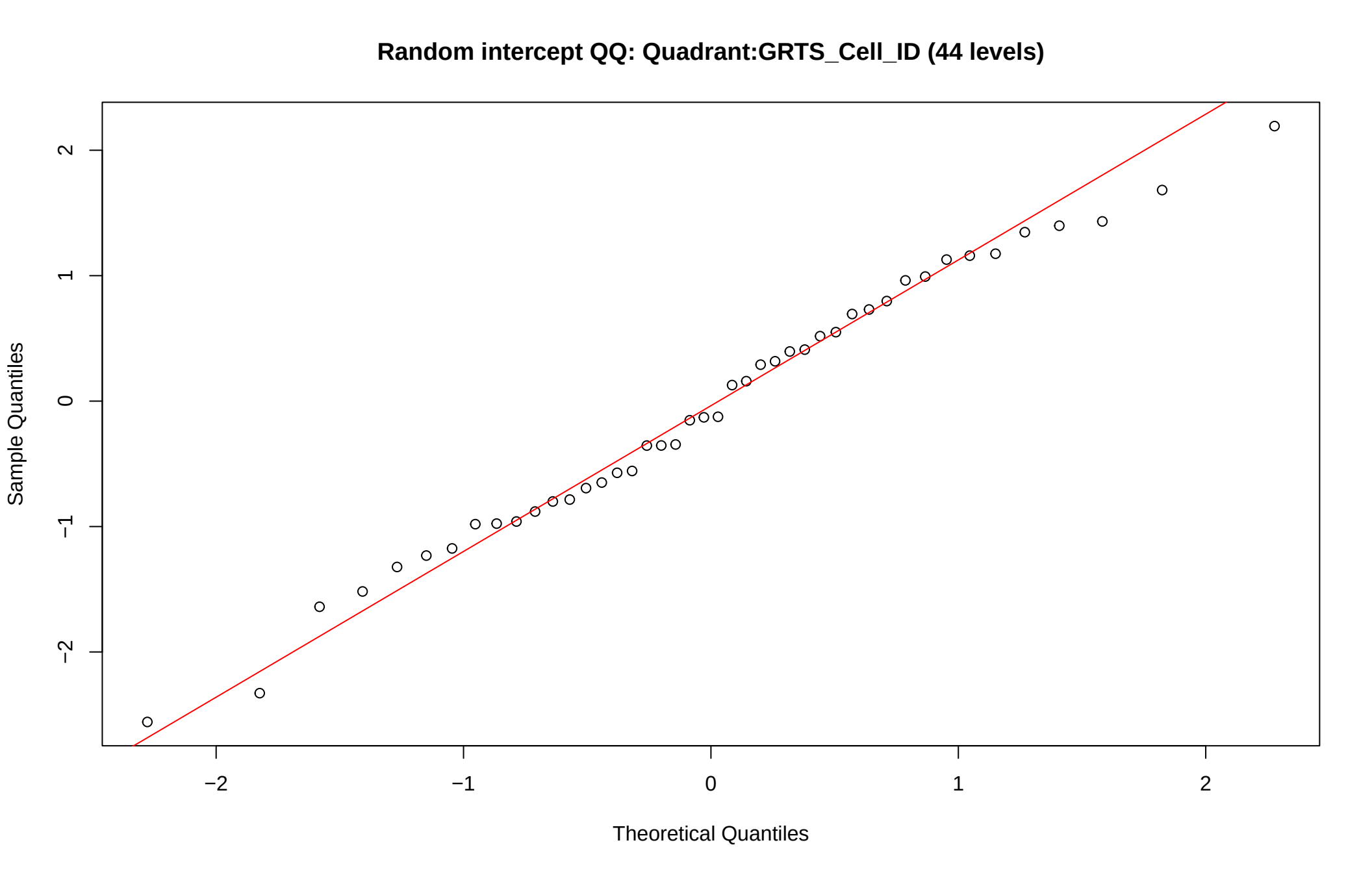

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
20
40
60
80
100
15
20
25
30
35
40
Simulated values, red line = fitted model. p−value (two.sided) = 0.052

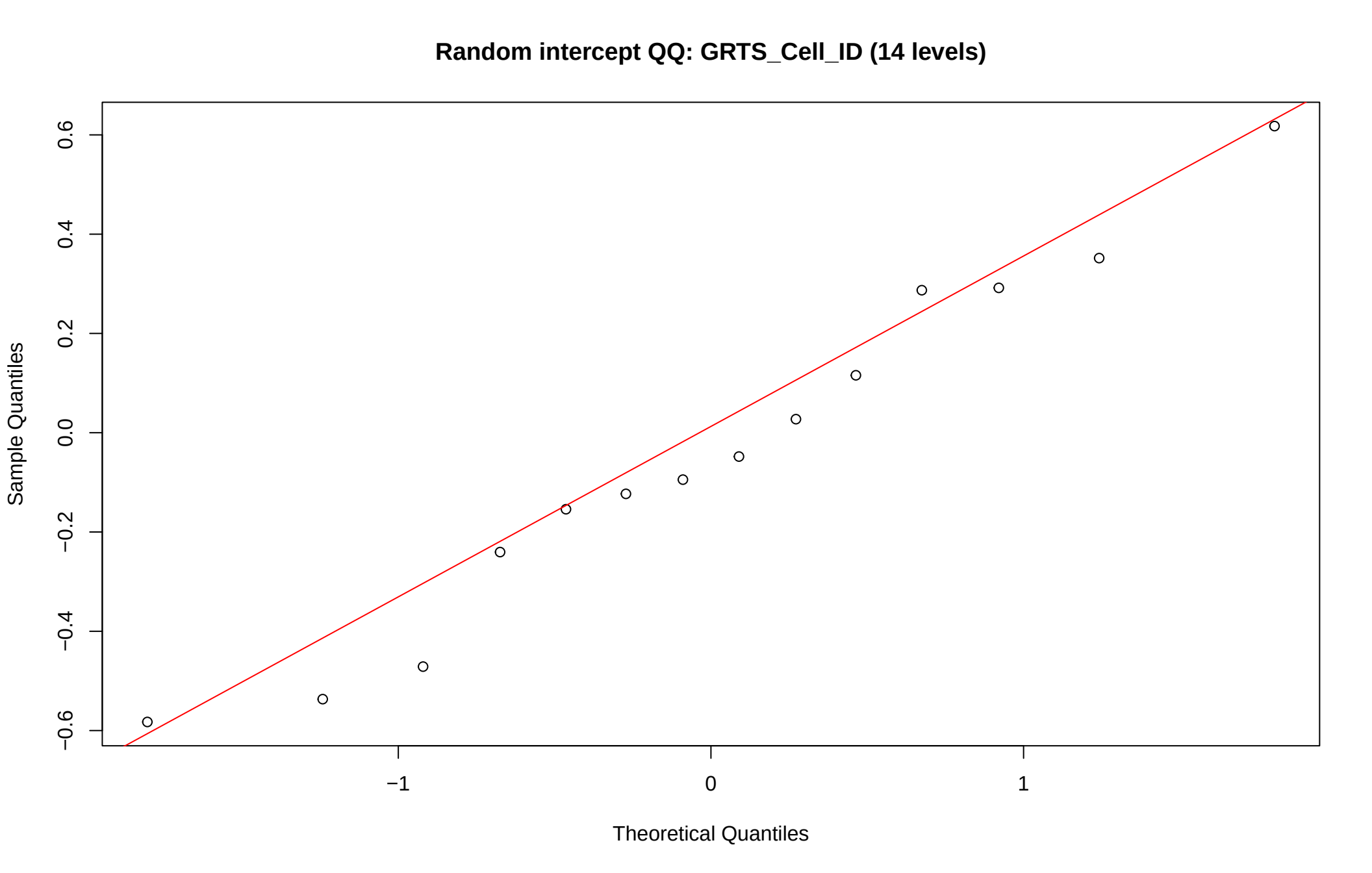

Residuals vs. time
Scaled residuals
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Time


Autocorrelation
Durbin−Watson test p= 0.22135
Deviation n.s.
ACF
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
0.00
0.25
0.50
0.75
1.00
115322
175354
186490
26170
283586
37274
47322
94522
986
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
115322_NW
175354_SW
186490_SW
26170_NE
269777_SW
294832_NW
37274_SW
94522_NE1
986_SE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (44 levels)
Sample Quantiles
Theoretical Quantiles
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (14 levels)
Sample Quantiles
Theoretical Quantiles
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−1
0
1